Matemáticas I
1º Bach.

SOLUCIONES DE LA FICHA · UNIDAD 5

Trigonometría

FentMates

Intenta resolver cada ejercicio antes de mirar su solución. Los resultados se dan de manera exacta cuando es posible y, si no, con dos decimales; los ángulos, en grados, minutos y segundos. Puedes comprobar los triángulos, las razones y las ecuaciones con las herramientas de los apuntes interactivos de la unidad.

1 Medida de ángulos

1.

a) $\boxed{\frac{5\pi}{4}}$ **b)** $\boxed{\frac{5\pi}{12}}$ **c)** $\boxed{\frac{8\pi}{9}}$ **d)** $\boxed{-85°56'37''}$

e) $\boxed{114°35'30''}$ **f)** $\boxed{3\pi}$ **g)** $\boxed{-\frac{3\pi}{2}}$ **h)** $\boxed{\frac{211\pi}{270}} \approx 2{,}4551$

i) $\boxed{60°}$ **j)** $\boxed{17°11'19''}$ **k)** $\boxed{-216°}$ **l)** $\boxed{\frac{7\pi}{4}}$

2 Trigonometría con la calculadora

2.

a) $\operatorname{sen}(319°12'52'') \approx \boxed{-0{,}6532}$ **b)** $\cos(434°26') \approx \boxed{0{,}2684}$ **c)** $\operatorname{tg}\frac{11\pi}{8} \approx \boxed{2{,}4142}$

d) $\sec 135° \approx \boxed{-1{,}4142}$ **e)** $\operatorname{tg} 7{,}03 \approx \boxed{0{,}9257}$ **f)** $\operatorname{cosec} 5{,}24 \approx \boxed{-1{,}1574}$

g) $\operatorname{sen}\frac{8\pi}{5} \approx \boxed{-0{,}9511}$ **h)** $\operatorname{cosec}(55'45'') \approx \boxed{61{,}6663}$ **i)** $\cos\frac{\pi}{3} \approx \boxed{0{,}5000}$

j) $\operatorname{cosec}(200°16') \approx \boxed{-2{,}8869}$ **k)** $\sec\frac{5\pi}{4} \approx \boxed{-1{,}4142}$ **l)** $\operatorname{tg}(183°13'53'') \approx \boxed{0{,}0565}$

m) $\operatorname{sen}(333°0'55'') \approx \boxed{-0{,}4538}$ **n)** $\cos 3{,}845 \approx \boxed{-0{,}7626}$ **ñ)** $\operatorname{cotg}\frac{11\pi}{6} \approx \boxed{-1{,}7321}$

3.

a) $\cos x$: $\boxed{63°0'21'', 296°59'39''}$ **b)** $\operatorname{sen} x$: $\boxed{68°11'12'', 111°48'48''}$

c) $\operatorname{tg} x$: $\boxed{151°22'12'', 331°22'12''}$ **d)** $\cos x$: $\boxed{113°33'11'', 246°26'49''}$

e) $\operatorname{cotg} x$: $\boxed{24°49'27'', 204°49'27''}$ **f)** $\cos x$: $\boxed{107°0'49'', 252°59'11''}$

g) $\operatorname{sen} x$: $\boxed{43°19'48'', 136°40'12''}$ **h)** $\operatorname{sen} x$: $\boxed{199°19'2'', 340°40'58''}$

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3 Razones trigonométricas de un ángulo agudo

4.

Grados	$0°$	$30°$	$45°$	$60°$	$90°$
Radianes	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sen	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tg	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\nexists$
cosec	$\nexists$	2	$\sqrt{2}$	$\frac{2\sqrt{3}}{3}$	1
sec	1	$\frac{2\sqrt{3}}{3}$	$\sqrt{2}$	2	$\nexists$
cotg	$\nexists$	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0

5.

a) α: $\operatorname{sen}\alpha = \boxed{\frac{2}{3}}$, $\operatorname{tg}\alpha = \boxed{\frac{2\sqrt{5}}{5}}$; $\sec\alpha = \frac{3\sqrt{5}}{5}$, $\operatorname{cosec}\alpha = \frac{3}{2}$, $\operatorname{cotg}\alpha = \frac{\sqrt{5}}{2}$

b) β: $\cos\beta = \boxed{\frac{4}{5}}$, $\operatorname{tg}\beta = \boxed{\frac{3}{4}}$; $\sec\beta = \frac{5}{4}$, $\operatorname{cosec}\beta = \frac{5}{3}$, $\operatorname{cotg}\beta = \frac{4}{3}$

c) γ: $\cos\gamma = \boxed{\frac{\sqrt{10}}{10}}$, $\operatorname{sen}\gamma = \boxed{\frac{3\sqrt{10}}{10}}$; $\sec\gamma = \sqrt{10}$, $\operatorname{cosec}\gamma = \frac{\sqrt{10}}{3}$, $\operatorname{cotg}\gamma = \frac{1}{3}$

d) α: $\operatorname{sen} \approx \boxed{0{,}5103}$, $\cos \approx \boxed{0{,}8600}$, $\operatorname{tg} \approx \boxed{0{,}5934}$

e) β: $\operatorname{sen} \approx \boxed{0{,}9429}$, $\cos \approx \boxed{0{,}3332}$, $\operatorname{tg} \approx \boxed{2{,}8300}$

f) $\gamma = 45°$: $\operatorname{sen}\gamma = \cos\gamma = \boxed{\frac{\sqrt{2}}{2}}$

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4 Razones trigonométricas de ángulos cualesquiera

6.

Grados	Radianes	sen	cos	tg
$120°$	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
$135°$	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
$150°$	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
$180°$	π	0	-1	0
$210°$	$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$225°$	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	1
$240°$	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
$270°$	$\frac{3\pi}{2}$	-1	0	$\nexists$
$300°$	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$
$315°$	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1
$330°$	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
$360°$	2π	0	1	0

7.

a) $\boxed{-2}$ **b)** $\boxed{1}$ **c)** $\boxed{-1}$ **d)** $\boxed{1}$

8.

Cuadrante	sen	cos	tg
Segundo	$0{,}6702$	$\boxed{-0{,}7422}$	$\boxed{-0{,}9030}$
Tercero	$\boxed{-0{,}8911}$	$-0{,}4539$	$\boxed{1{,}9631}$
Cuarto	$\boxed{-0{,}5979}$	$\boxed{0{,}8016}$	$-0{,}7459$
Tercero	$-0{,}7822$	$\boxed{-0{,}6230}$	$\boxed{1{,}2555}$
Segundo	$\boxed{0{,}8850}$	$\boxed{-0{,}4657}$	$-1{,}9004$
Cuarto	$\boxed{-0{,}7158}$	$0{,}6983$	$\boxed{-1{,}0251}$

9.

a) $\cos\alpha = \frac{1}{4}$, $\operatorname{sen}\alpha = \boxed{-\frac{\sqrt{15}}{4}}$ (negativo en el cuarto cuadrante), $\operatorname{tg}\alpha = \boxed{-\sqrt{15}}$

b) $\operatorname{tg}\alpha = -2$, $\cos\alpha = -\frac{1}{\sqrt{1+4}} = \boxed{-\frac{\sqrt{5}}{5}}$, $\operatorname{sen}\alpha = \operatorname{tg}\alpha \cdot \cos\alpha = \boxed{\frac{2\sqrt{5}}{5}}$

10. $\operatorname{sen}\alpha \approx \boxed{0{,}62}$, $\cos\alpha \approx \boxed{-0{,}78}$, $\operatorname{tg}\alpha \approx \boxed{-0{,}79}$, $\operatorname{cosec}\alpha \approx \boxed{1{,}61}$, $\sec\alpha \approx \boxed{-1{,}27}$, $\operatorname{cotg}\alpha \approx \boxed{-1{,}27}$

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11. $\operatorname{sen}\alpha \approx \boxed{-0{,}56}$, $\cos\alpha \approx \boxed{-0{,}83}$, $\operatorname{tg}\alpha \approx \boxed{0{,}67}$, $\operatorname{cosec}\alpha \approx \boxed{-1{,}79}$, $\sec\alpha \approx \boxed{-1{,}20}$, $\operatorname{cotg}\alpha \approx \boxed{1{,}49}$

12. $\operatorname{sen}\alpha \approx \boxed{-0{,}68}$, $\cos\alpha \approx \boxed{0{,}74}$, $\operatorname{tg}\alpha \approx \boxed{-0{,}92}$, $\operatorname{cosec}\alpha \approx \boxed{-1{,}48}$, $\sec\alpha \approx \boxed{1{,}36}$, $\operatorname{cotg}\alpha \approx \boxed{-1{,}09}$

13.

a) $\operatorname{sen} 13290° = \operatorname{sen} 330° = \boxed{-\frac{1}{2}}$

b) $\cos(-1680°) = \cos 120° = \boxed{-\frac{1}{2}}$

c) $\operatorname{tg} 3825° = \operatorname{tg} 225° = \boxed{1}$

d) $\cos 4995° = \cos 315° = \boxed{\frac{\sqrt{2}}{2}}$

e) $\operatorname{sen}(-1710°) = \operatorname{sen} 90° = \boxed{1}$

f) $\operatorname{tg} 3630° = \operatorname{tg} 30° = \boxed{\frac{\sqrt{3}}{3}}$

g) $\cos(-36000°) = \cos 0° = \boxed{1}$

h) $\operatorname{sen}(-330°) = \operatorname{sen} 30° = \boxed{\frac{1}{2}}$

5 Relaciones entre las razones trigonométricas de algunos ángulos

14.

a) $\cos\alpha = \boxed{-\frac{2}{3}}$, $\operatorname{sen}\alpha = \boxed{\frac{\sqrt{5}}{3}}$, $\operatorname{sen}(180° - \alpha) = \operatorname{sen}\alpha = \boxed{\frac{\sqrt{5}}{3}}$, $\cos(90° + \alpha) = -\operatorname{sen}\alpha = \boxed{-\frac{\sqrt{5}}{3}}$,

$\operatorname{tg}(180° + \alpha) = \operatorname{tg}\alpha = \boxed{-\frac{\sqrt{5}}{2}}$

b) $\alpha = 180° - \operatorname{tg}^{-1}\frac{\sqrt{5}}{2} \approx \boxed{131°48'37''}$

15.

a) $\boxed{-0{,}8}$

b) $\boxed{0{,}6}$

c) $\boxed{-0{,}75}$

d) $\boxed{0{,}8}$

e) $\boxed{-0{,}6}$

f) $\boxed{\frac{4}{3}}$

16.

a) $\cos 48° = \operatorname{sen} 42° = \boxed{0{,}67}$

b) $\operatorname{sen}(-48°) = -\cos 42° = \boxed{-0{,}74}$

c) $\operatorname{sen} 138° = \operatorname{sen} 42° = \boxed{0{,}67}$

d) $\operatorname{tg} 318° = -\operatorname{tg} 42° = \boxed{-0{,}9}$

e) $\cos 222° = -\cos 42° = \boxed{-0{,}74}$

f) $\operatorname{tg} 858° = \operatorname{tg} 138° = -\operatorname{tg} 42° = \boxed{-0{,}9}$

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6 Resolución de triángulos rectángulos

17.

a) $a = c\cos\hat{B} \approx \boxed{17{,}43 \text{ cm}}$, $b = c\,\text{sen}\,\hat{B} \approx \boxed{26{,}84 \text{ cm}}$ $(\hat{A} = 33°)$

b) $c = \sqrt{250^2 + 308^2} \approx \boxed{396{,}69 \text{ m}}$, $\text{tg}\,\hat{A} = \frac{250}{308} \Rightarrow \hat{A} \approx \boxed{39°3'57''}$

c) $b = \dfrac{a}{\text{tg}\,\hat{A}} \approx \boxed{56{,}01 \text{ cm}}$, $c = \dfrac{a}{\text{sen}\,\hat{A}} \approx \boxed{66{,}05 \text{ cm}}$ $(\hat{B} = 58°)$

18. Altura: $12\,\text{sen}\,50° \approx \boxed{9{,}19 \text{ m}}$. Distancia: $12\cos 50° \approx \boxed{7{,}71 \text{ m}}$.

19. Con A y B a la misma altura, como en la figura, sumamos las proyecciones horizontales de los cuatro tramos: $4\cos 60° + 1{,}7\cos 30° + 3{,}5\cos 20° + 4{,}2\cos 50° \approx 2 + 1{,}47 + 3{,}29 + 2{,}7 \approx \boxed{9{,}46 \text{ m}}$

20. Altura a los 6 km: $6\,\text{tg}\,15° \approx 1{,}608$ km. A los 11 km: $1{,}608 + 5\,\text{tg}\,25° \approx \boxed{3{,}939 \text{ km}} > 2{,}5$ km: $\boxed{\text{pasa por encima de la montaña}}$.

21. La altura lo divide en dos triángulos rectángulos con un cateto de 5 m y un ángulo de $20°$: lado igual $= \dfrac{5}{\text{sen}\,20°} \approx 14{,}62$ m, altura $= \dfrac{5}{\text{tg}\,20°} \approx 13{,}74$ m. Perímetro $\approx \boxed{39{,}24 \text{ m}}$, área $\approx \boxed{68{,}69 \text{ m}^2}$.

22. $S = 9 \cdot 16 \cdot \text{sen}\,65° \approx \boxed{130{,}51 \text{ cm}^2}$

7 Teorema del seno y teorema del coseno

23.

a) Teorema del coseno: $\hat{A} \approx \boxed{44°24'55''}$, $\hat{B} \approx \boxed{101°32'13''}$, $\hat{C} \approx \boxed{34°2'52''}$

b) Teorema del coseno: $a \approx \boxed{5{,}77 \text{ cm}}$; $\hat{B} \approx \boxed{41°4'43''}$, $\hat{C} \approx \boxed{99°43'17''}$

c) $\hat{A} = \boxed{74°31'}$; teorema del seno: $b \approx \boxed{4{,}55 \text{ cm}}$, $c \approx \boxed{6{,}67 \text{ cm}}$

d) Teorema del seno: $\text{sen}\,\hat{A} \approx 0{,}4774 \Rightarrow \hat{A} \approx \boxed{28°30'46''}$ (el obtuso no cabe); $\hat{C} \approx \boxed{114°51'14''}$, $c \approx \boxed{15{,}21 \text{ cm}}$

e) Teorema del coseno: $b \approx \boxed{11{,}27 \text{ cm}}$; $\hat{A} \approx \boxed{50°58'46''}$, $\hat{C} \approx \boxed{25°34'14''}$

24.

a) $12^2 = 144 < 9^2 + 11^2 = 202$: $\boxed{\text{acutángulo}}$

b) $\boxed{\text{no es un triángulo}}$ $(23 \geq 8 + 14)$

c) $26^2 = 676 = 10^2 + 24^2 = 676$: $\boxed{\text{rectángulo}}$

d) $40^2 = 1600 > 20^2 + 30^2 = 1300$: $\boxed{\text{obtusángulo}}$

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8 Resolución de triángulos: problemas

25. $8 \operatorname{sen} 38° \approx \boxed{4{,}93 \text{ m}}$

26. $40 \operatorname{tg} 26° + 1{,}75 \approx 19{,}51 + 1{,}75 \approx \boxed{21{,}26 \text{ m}}$

27. Teorema del coseno: $\cos\theta = \dfrac{6^2 + 12^2 - 8^2}{2 \cdot 6 \cdot 12} = \dfrac{116}{144} \Rightarrow \theta \approx \boxed{36°20'10''}$

28. $h = x \operatorname{tg} 40° = (x + 30) \operatorname{tg} 28° \Rightarrow x = \dfrac{30 \operatorname{tg} 28°}{\operatorname{tg} 40° - \operatorname{tg} 28°} \approx \boxed{51{,}89 \text{ m}}$, $h \approx \boxed{43{,}54 \text{ m}}$

29. La visual al árbol cercano forma con la vertical un ángulo α con $\operatorname{tg}\alpha = \frac{50}{25} = 2$, es decir, $\alpha \approx 63°26'6''$. La del árbol lejano forma $\alpha + 15°$: $50 + x = 25 \operatorname{tg}(\alpha + 15°) \Rightarrow x \approx \boxed{72{,}17 \text{ m}}$

30. $d = \dfrac{40}{\operatorname{sen} 15°} \approx \boxed{154{,}55 \text{ m}}$

31. $\dfrac{h}{\operatorname{tg} 43°} + \dfrac{h}{\operatorname{tg} 57°50'} = 15 \Rightarrow h \approx \boxed{8{,}82 \text{ m}}$

32.

a) $\boxed{30°, 60° \text{ y } 90°}$

b) Catetos x y $x\sqrt{3}$: $\frac{x^2\sqrt{3}}{2} = 28 \Rightarrow x \approx \boxed{5{,}69}$, el otro cateto $\approx \boxed{9{,}85}$, hipotenusa $\approx \boxed{11{,}37}$ cm; perímetro $\approx \boxed{26{,}91 \text{ cm}}$

33. $\operatorname{tg}\alpha = \frac{1{,}70}{0{,}50} = 3{,}4 \Rightarrow \alpha \approx \boxed{73°36'38''}$

34. La anchura del tejado es $\dfrac{8}{\cos 18°} \approx 8{,}41$ m: área $\approx 12 \cdot 8{,}41 \approx \boxed{100{,}94 \text{ m}^2}$

35. Después de 2 h han recorrido 10 y 8 km: $d = \sqrt{10^2 + 8^2 - 2 \cdot 10 \cdot 8 \cos 54°} \approx \boxed{8{,}36 \text{ km}}$. Después de 6 h (el triple): $\approx \boxed{25{,}09 \text{ km}}$.

36.

a) El tercer ángulo es $93°$. Teorema del seno: tramos de $\dfrac{8 \operatorname{sen} 37°}{\operatorname{sen} 93°} \approx 4{,}82$ m y $\dfrac{8 \operatorname{sen} 50°}{\operatorname{sen} 93°} \approx 6{,}14$ m: cuerda $\approx \boxed{10{,}96 \text{ m}}$

b) $d = 4{,}82 \cdot \operatorname{sen} 50° \approx \boxed{3{,}69 \text{ m}}$

37. Altura $h = 8 \operatorname{sen} 65° \approx 7{,}25$; proyección del lado $8 \cos 65° \approx 3{,}38$. Por Pitágoras con la diagonal, la base mayor es $B \approx 13{,}13 + 3{,}38 \approx 16{,}51$ y la menor $b \approx 9{,}75$ cm. Área $= \frac{B+b}{2} h \approx \boxed{95{,}21 \text{ cm}^2}$

38. Proyección del lado: $\frac{14-8}{2} = 3$ cm: $\cos\alpha = \frac{3}{5} \Rightarrow \alpha \approx \boxed{53°7'48''}$ (dos ángulos) y $180° - \alpha \approx \boxed{126°52'12''}$ (los otros dos)

39. $2 \operatorname{tg}^{-1} \frac{6}{10} \approx \boxed{61°55'39''}$ (y el suplementario, $\boxed{118°4'21''}$)

40. $\cos\theta = \dfrac{10^2 + 10^2 - 15^2}{2 \cdot 10 \cdot 10} = -0{,}125 \Rightarrow \theta \approx \boxed{97°10'51''}$; los otros dos: $\boxed{41°24'35''}$

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41. $\cos\alpha = \dfrac{5^2+8^2-10^2}{2\cdot 5\cdot 8} = -\frac{11}{80} \Rightarrow \alpha \approx \boxed{97°54'12''}$ (dos ángulos) y $\boxed{82°5'48''}$ (los otros dos)

9 Fórmulas trigonométricas

42. $15° = 45° - 30°$: $\operatorname{sen} = \boxed{\dfrac{\sqrt{6}-\sqrt{2}}{4}}$, $\cos = \boxed{\dfrac{\sqrt{6}+\sqrt{2}}{4}}$, $\operatorname{tg} = \boxed{2-\sqrt{3}}$. $\frac{5\pi}{12} = 75° = 45° + 30°$: $\operatorname{sen} = \boxed{\dfrac{\sqrt{6}+\sqrt{2}}{4}}$, $\cos = \boxed{\dfrac{\sqrt{6}-\sqrt{2}}{4}}$, $\operatorname{tg} = \boxed{2+\sqrt{3}}$.

43. $\cos\alpha = 0{,}8$, $\cos\beta = 0{,}6$.

a) $\boxed{1}$ **b)** $\boxed{-0{,}28}$ **c)** $\boxed{0}$

d) $\boxed{0{,}96}$ **e)** $\boxed{\nexists}$ $(\cos = 0)$ **f)** $\boxed{-\dfrac{7}{24}}$

44. $\operatorname{sen}\alpha\cos\frac{3\pi}{2} + \cos\alpha\operatorname{sen}\frac{3\pi}{2} = \operatorname{sen}\alpha\cdot 0 + \cos\alpha\cdot(-1) = \boxed{-\cos\alpha}$

45. $\operatorname{sen} 10° = \sqrt{\frac{1-0{,}94}{2}} \approx \boxed{0{,}1732}$, $\cos 10° = \sqrt{\frac{1+0{,}94}{2}} \approx \boxed{0{,}9849}$; $\operatorname{sen} 40° = 2\cdot 0{,}34\cdot 0{,}94 \approx \boxed{0{,}6392}$, $\cos 40° = 0{,}94^2 - 0{,}34^2 = \boxed{0{,}768}$

46. $\cos\alpha = -\frac{4}{5}$; $45° < \frac{\alpha}{2} < 90°$: $\operatorname{sen}\frac{\alpha}{2} = \boxed{\dfrac{3\sqrt{10}}{10}}$, $\cos\frac{\alpha}{2} = \boxed{\dfrac{\sqrt{10}}{10}}$, $\operatorname{tg}\frac{\alpha}{2} = \boxed{3}$

47. $\cos\alpha = -\frac{4}{5}$, $\operatorname{tg}\alpha = -\frac{3}{4}$, $\operatorname{sen}\beta = -\frac{\sqrt{15}}{4}$. $\cos(\alpha+\beta) = \boxed{\dfrac{4+3\sqrt{15}}{20}}$, $\operatorname{sen}(\alpha-\beta) = \boxed{-\dfrac{3+4\sqrt{15}}{20}}$, $\operatorname{tg} 2\alpha = \boxed{-\dfrac{24}{7}}$, $\operatorname{tg}\frac{\beta}{2} = \boxed{-\dfrac{\sqrt{15}}{3}}$ ($\frac{\beta}{2}$ es del segundo cuadrante)

48. En los dos casos: $\operatorname{sen} 2\alpha = \boxed{\dfrac{3}{5}}$, $\cos 2\alpha = \boxed{-\dfrac{4}{5}}$, $\operatorname{tg} 2\alpha = \boxed{-\dfrac{3}{4}}$ (los dos valores de 2α difieren en $360°$).

49. $45° < \frac{\alpha}{2} < 90°$: $\operatorname{sen}\frac{\alpha}{2} = \sqrt{\frac{5}{6}} = \boxed{\dfrac{\sqrt{30}}{6}}$, $\cos\frac{\alpha}{2} = \sqrt{\frac{1}{6}} = \boxed{\dfrac{\sqrt{6}}{6}}$, $\operatorname{tg}\frac{\alpha}{2} = \boxed{\sqrt{5}}$

50.

a) $\dfrac{\operatorname{sen}\alpha - \cos\alpha}{\frac{\operatorname{sen}\alpha-\cos\alpha}{\cos\alpha}} = \cos\alpha$

b) $\dfrac{\operatorname{sen}^2\alpha - \operatorname{sen}^2\alpha\cos^2\alpha}{\cos^2\alpha} = \dfrac{\operatorname{sen}^2\alpha(1-\cos^2\alpha)}{\cos^2\alpha} = \operatorname{tg}^2\alpha\operatorname{sen}^2\alpha$

c) $\dfrac{\frac{\operatorname{sen}\alpha+\cos\alpha}{\operatorname{sen}\alpha}}{\operatorname{sen}\alpha+\cos\alpha} = \dfrac{1}{\operatorname{sen}\alpha}$

d) $\dfrac{1-\cos^2\alpha}{\cos^2\alpha} = \dfrac{\operatorname{sen}^2\alpha}{\cos^2\alpha}$

e) $\dfrac{\operatorname{sen}^2\alpha + \cos^2\alpha + 2\operatorname{sen}\alpha\cos\alpha}{2\cos^2\alpha} = \dfrac{(\operatorname{sen}\alpha+\cos\alpha)^2}{2\cos^2\alpha} = \frac{1}{2}(\operatorname{tg}\alpha+1)^2$

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f) $2\,\text{tg}\,\alpha \cdot \dfrac{1-\cos\alpha}{2} + \operatorname{sen}\alpha = \text{tg}\,\alpha - \operatorname{sen}\alpha + \operatorname{sen}\alpha$

g) $\cos 2\alpha\cos\alpha - \operatorname{sen}2\alpha\operatorname{sen}\alpha = (2\cos^2\alpha - 1)\cos\alpha - 2(1-\cos^2\alpha)\cos\alpha$

h) $\dfrac{\operatorname{sen}2\alpha}{\cos 2\alpha} - \dfrac{\operatorname{sen}\alpha}{\cos\alpha} = \dfrac{\operatorname{sen}(2\alpha-\alpha)}{\cos 2\alpha\cos\alpha} = \dfrac{\text{tg}\,\alpha}{\cos 2\alpha}$

i) $\dfrac{\operatorname{sen}^2\alpha + \cos^2\alpha}{\operatorname{sen}\alpha\cos\alpha} = \dfrac{1}{\cos\alpha}\cdot\dfrac{1}{\operatorname{sen}\alpha}$

j) $\dfrac{2\operatorname{sen}\alpha\cos\alpha}{2\cos^2\alpha} = \dfrac{\operatorname{sen}\alpha}{\cos\alpha}$

k) $\cos^2\alpha + \dfrac{\cos^4\alpha}{\operatorname{sen}^2\alpha} = \dfrac{\cos^2\alpha(\operatorname{sen}^2\alpha+\cos^2\alpha)}{\operatorname{sen}^2\alpha} = \dfrac{1}{\text{tg}^2\alpha}$

51.

a) Es $\cos(\alpha-(\alpha-\beta)) = \cos\beta$.

b) $\operatorname{sen}^2\alpha\cos^2\beta - \cos^2\alpha\operatorname{sen}^2\beta = \operatorname{sen}^2\alpha(1-\operatorname{sen}^2\beta) - (1-\operatorname{sen}^2\alpha)\operatorname{sen}^2\beta$

c) $\dfrac{2\cos\alpha\cos\beta}{2\operatorname{sen}\alpha\cos\beta}$

d) $2 - 2(\cos\alpha\cos\beta - \operatorname{sen}\alpha\operatorname{sen}\beta) = 2 - 2\cos(\alpha+\beta) = 4\operatorname{sen}^2\frac{\alpha+\beta}{2}$

10 Ecuaciones trigonométricas

52.

a) $\operatorname{sen}x = \frac{1}{2}$: $\boxed{30°}$, $\boxed{150°}$ $(x = 30° + 360°k,\ 150° + 360°k)$

b) $\cos x = \frac{\sqrt{3}}{2}$: $\boxed{30°}$, $\boxed{330°}$ $(x = \pm 30° + 360°k)$

c) $\text{tg}\,x = 1$: $\boxed{45°}$, $\boxed{225°}$ $(x = 45° + 180°k)$

d) $\operatorname{sen}x = -\frac{\sqrt{2}}{2}$: $\boxed{225°}$, $\boxed{315°}$ $(x = 225° + 360°k,\ 315° + 360°k)$

e) $\cos x = -\frac{1}{2}$: $\boxed{120°}$, $\boxed{240°}$ $(x = \pm 120° + 360°k)$

f) $1 + \cos x = 0$: $\boxed{180°}$ $(x = 180° + 360°k)$

g) $\operatorname{sen}4x = -\frac{\sqrt{3}}{2}$: $\boxed{60°}$, $\boxed{75°}$, $\boxed{150°}$, $\boxed{165°}$, $\boxed{240°}$, $\boxed{255°}$, $\boxed{330°}$, $\boxed{345°}$ $(x = 60° + 90°k,\ 75° + 90°k)$

h) $\cos 2x = \frac{\sqrt{2}}{2}$: $\boxed{22°30'}$, $\boxed{157°30'}$, $\boxed{202°30'}$, $\boxed{337°30'}$ $(x = \pm 22°30' + 180°k)$

i) $\text{tg}\,3x = -1$: $\boxed{45°}$, $\boxed{105°}$, $\boxed{165°}$, $\boxed{225°}$, $\boxed{285°}$, $\boxed{345°}$ $(x = 45° + 60°k)$

j) $\operatorname{sen}\frac{x}{2} = 0$: $\boxed{0°}$ $(x = 360°k)$

k) $\cos\frac{x}{3} = -\frac{1}{2}$: $\boxed{\text{ninguna}}$ en $[0°,360°)$ $(x = 360° + 1080°k,\ 720° + 1080°k)$

l) $\text{tg}\,\frac{3x}{4} = -\frac{\sqrt{3}}{3}$: $\boxed{200°}$ $(x = 200° + 240°k)$

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53.

a) $\cos(2x+30°) = \pm\frac{\sqrt{2}}{2} \Rightarrow 2x+30° = 45°+90°k \Rightarrow x = 7°30'+45°k$: $\boxed{7°30'}$, $\boxed{52°30'}$, $\boxed{97°30'}$, $\boxed{142°30'}$, $\boxed{187°30'}$, $\boxed{232°30'}$, $\boxed{277°30'}$, $\boxed{322°30'}$

b) $\cos x(\cos x - \sqrt{3}\operatorname{sen} x) = 0$: $\cos x = 0$ o $\operatorname{tg} x = \frac{\sqrt{3}}{3}$: $\boxed{30°}$, $\boxed{90°}$, $\boxed{210°}$, $\boxed{270°}$

c) $2\cos(x+30°) = 1 \Rightarrow x+30° = 60°$ o $300°$: $\boxed{30°}$, $\boxed{270°}$

d) $\operatorname{sen}^2 x = -2\cos x \Rightarrow \cos^2 x - 2\cos x - 1 = 0 \Rightarrow \cos x = 1-\sqrt{2}$: $\boxed{114°28'11''}$, $\boxed{245°31'49''}$

e) $\frac{2\operatorname{sen} 2x\cos x}{2\cos 2x\cos x} = \operatorname{tg} 2x = \sqrt{3}$: $\boxed{30°}$, $\boxed{120°}$, $\boxed{210°}$, $\boxed{300°}$

f) $\operatorname{sen} x(\operatorname{sen}^2 x - \cos^2 x) = -\operatorname{sen} x\cos 2x = 0$: $\boxed{0°}$, $\boxed{45°}$, $\boxed{135°}$, $\boxed{180°}$, $\boxed{225°}$, $\boxed{315°}$

g) $2\operatorname{sen}(x+30°) = 2 \Rightarrow x+30° = 90°$: $\boxed{60°}$

h) $\frac{1-\cos x}{1+\cos x} = 1-\cos x \Rightarrow \cos x = 1$ o $\cos x = 0$: $\boxed{0°}$, $\boxed{90°}$, $\boxed{270°}$

i) $\frac{2\cos 3x\cos x}{2\cos 3x\operatorname{sen} x} = \operatorname{cotg} x = 1$: $\boxed{45°}$, $\boxed{225°}$

j) Dividiendo entre $\operatorname{sen} x \neq 0$: $4\cos^2 x + 4\cos x + 1 = 0 \Rightarrow \cos x = -\frac{1}{2}$: $\boxed{120°}$, $\boxed{240°}$

54.

a) $\operatorname{tg}^2 x - 4\operatorname{tg} x + 3 = 0 \Rightarrow \operatorname{tg} x = 1$ o 3: $\boxed{45°}$, $\boxed{71°33'54''}$, $\boxed{225°}$, $\boxed{251°33'54''}$

b) $2\operatorname{sen} x(\cos^2 x - 3\operatorname{sen}^2 x) = 0$: $\operatorname{sen} x = 0$ o $\operatorname{tg}^2 x = \frac{1}{3}$: $\boxed{0°}$, $\boxed{30°}$, $\boxed{150°}$, $\boxed{180°}$, $\boxed{210°}$, $\boxed{330°}$

c) $8\operatorname{sen}^2 x + 2\operatorname{sen} x - 3 = 0 \Rightarrow \operatorname{sen} x = \frac{1}{2}$ o $-\frac{3}{4}$: $\boxed{30°}$, $\boxed{150°}$, $\boxed{228°35'25''}$, $\boxed{311°24'35''}$

d) $\operatorname{tg} x = \frac{\sqrt{3}}{3}$: $\boxed{30°}$, $\boxed{210°}$

e) $\operatorname{sen} x(2\cos^2 x - 1) = 0$: $\boxed{0°}$, $\boxed{45°}$, $\boxed{135°}$, $\boxed{180°}$, $\boxed{225°}$, $\boxed{315°}$

f) $2\cos 2x\cos x = 0$: $\boxed{45°}$, $\boxed{90°}$, $\boxed{135°}$, $\boxed{225°}$, $\boxed{270°}$, $\boxed{315°}$

g) $2\cos^2 x - \cos x - 3 = 0 \Rightarrow \cos x = -1$ ($\frac{3}{2}$ no vale): $\boxed{180°}$

h) $\sqrt{2}\operatorname{sen}^2 x + \operatorname{sen} x - \sqrt{2} = 0 \Rightarrow \operatorname{sen} x = \frac{\sqrt{2}}{2}$: $\boxed{45°}$, $\boxed{135°}$

i) $2\cos 2x = 7\operatorname{sen} x \Rightarrow 4\operatorname{sen}^2 x + 7\operatorname{sen} x - 2 = 0 \Rightarrow \operatorname{sen} x = \frac{1}{4}$ (-2 no vale): $\boxed{14°28'39''}$, $\boxed{165°31'21''}$

j) $4\cos^2 x + 8\cos x - 5 = 0 \Rightarrow \cos x = \frac{1}{2}$ ($-\frac{5}{2}$ no vale): $\boxed{60°}$, $\boxed{300°}$

k) $-2\operatorname{sen}^2 x - 4\operatorname{sen} x = 0 \Rightarrow \operatorname{sen} x = 0$ (-2 no vale): $\boxed{0°}$, $\boxed{180°}$

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11 Sistemas de ecuaciones trigonométricas

55.

a) $\cos y = 1 + \operatorname{sen} x \Rightarrow (2\operatorname{sen} x + 1)^2 = 0$: $\operatorname{sen} x = -\frac{1}{2}$, $\cos y = \frac{1}{2}$: $\boxed{(210°,60°)}$, $\boxed{(210°,300°)}$, $\boxed{(330°,60°)}$, $\boxed{(330°,300°)}$

b) $\cos x$ y $\cos y$ son las raíces de $t^2 + \frac{1}{2}t - \frac{1}{2} = 0$: $\frac{1}{2}$ y -1: $\boxed{(60°,180°)}$, $\boxed{(180°,60°)}$, $\boxed{(180°,300°)}$, $\boxed{(300°,180°)}$

c) $\operatorname{sen} x \operatorname{sen} y = \frac{(\operatorname{sen} x+\operatorname{sen} y)^2-(\operatorname{sen}^2 x+\operatorname{sen}^2 y)}{2} = 0$: un seno vale 0 y el otro $\frac{\sqrt{3}}{2}$: $\boxed{(0°,60°)}$, $\boxed{(0°,120°)}$, $\boxed{(60°,0°)}$, $\boxed{(60°,180°)}$, $\boxed{(120°,0°)}$, $\boxed{(120°,180°)}$, $\boxed{(180°,60°)}$, $\boxed{(180°,120°)}$

d) $\cos y = \operatorname{sen} x \Rightarrow \sqrt{3}\cos x - \operatorname{sen} x = 1 \Rightarrow \cos(x + 30°) = \frac{1}{2}$: $\boxed{x = \frac{\pi}{6}, y = \frac{\pi}{3}}$ (con $x = \frac{3\pi}{2}$, $y = -\pi$ quedaría fuera del intervalo)

e) Sumando y restando: $\operatorname{sen}(x+y) = \frac{1}{2}$, $\operatorname{sen}(x-y) = 0$: $\boxed{(15°,15°)}$, $\boxed{(75°,75°)}$, $\boxed{(105°,285°)}$, $\boxed{(165°,345°)}$, $\boxed{(195°,195°)}$, $\boxed{(255°,255°)}$, $\boxed{(285°,105°)}$, $\boxed{(345°,165°)}$

56.

a) $y = \frac{\pi}{2} - x \Rightarrow \operatorname{sen} x + \cos x = 1$: $\boxed{(0,\frac{\pi}{2})}$ y $\boxed{(\frac{\pi}{2},0)}$

b) $\operatorname{sen}\left(y + \frac{\pi}{3}\right) = 2\operatorname{sen} y \Rightarrow \operatorname{tg} y = \frac{\sqrt{3}}{3}$: $\boxed{y = \frac{\pi}{6}, x = \frac{\pi}{2}}$ y $\boxed{y = \frac{7\pi}{6}, x = \frac{3\pi}{2}}$

c) $2y \operatorname{sen} 2x = 3$ y $2y\cos 2x = \sqrt{3}$: dividiendo, $\operatorname{tg} 2x = \sqrt{3}$; sumando los cuadrados, $4y^2 = 12$. En $[0,2\pi)$: $\boxed{x = \frac{\pi}{6}, \frac{7\pi}{6} \text{ con } y = \sqrt{3}}$ y $\boxed{x = \frac{2\pi}{3}, \frac{5\pi}{3} \text{ con } y = -\sqrt{3}}$

12 Funciones trigonométricas

57.

a) $\boxed{\text{V}}$ **b)** $\boxed{\text{V}}$ **c)** $\boxed{\text{V}}$ **d)** $\boxed{\text{V}}$ ($\cos x = \operatorname{sen}(x + \frac{\pi}{2})$)

58.

a) $\frac{x}{2} = \frac{\pi}{2} + k\pi$: $\boxed{x = \pi,\ 3\pi}$

b) $x - \pi = k\pi$: $\boxed{x = 0,\ \pi,\ 2\pi,\ 3\pi,\ 4\pi}$

c) $x + \pi = \frac{\pi}{2} + k\pi$: $\boxed{x = \frac{\pi}{2},\ \frac{3\pi}{2},\ \frac{5\pi}{2},\ \frac{7\pi}{2}}$

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59.

a) $= \cos x$: la gráfica del coseno

b) $= -\operatorname{sen} x$: la del seno reflejada en el eje X

c) $= \cos x$: la gráfica del coseno

d) $= \operatorname{sen} x$: la gráfica del seno

60.

a) a) $y = 2\cos x \to (3)$ **b)** b) $y = \cos 2x \to (7)$ **c)** c) $y = \frac{1}{2}\operatorname{sen} x \to (8)$ **d)** d) $y = \cos\frac{x}{2} \to (2)$

e) e) $y = \operatorname{sen} 2x \to (6)$ **f)** f) $y = 2\operatorname{sen} x \to (1)$ **g)** g) $y = \frac{1}{2}\cos x \to (5)$ **h)** h) $y = \operatorname{sen}\frac{x}{2} \to (4)$

Fíjate en la amplitud (el valor máximo) y en el período: 2π, π (si hay $2x$) o 4π (si hay $\frac{x}{2}$).

13 Ampliación y problemas (ejercicios nuevos)

61. 25 min $= 150° = \frac{5\pi}{6}$ rad: $l = 12 \cdot \frac{5\pi}{6} =$ $10\pi \approx 31{,}42$ cm; $S = \frac{1}{2} \cdot 144 \cdot \frac{5\pi}{6} =$ $60\pi \approx 188{,}5$ cm^2

62. $\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{3} = \frac{\sqrt{6}}{4} + \frac{\sqrt{3}}{3}$; $\frac{\frac{1}{2} + \frac{1}{2}}{-1} = -1$

63. $\operatorname{sen} x - \operatorname{sen} x - \operatorname{sen} x = -\operatorname{sen} x$; $\frac{(-\cos x)(-\operatorname{tg} x)}{-\operatorname{sen} x} = \frac{\operatorname{sen} x}{-\operatorname{sen} x} = -1$

64. $\operatorname{sen}\hat{B} = \frac{8\operatorname{sen} 40°}{6} \approx 0{,}8571$: dos soluciones. $\hat{B}_1 \approx 58°59'13''$, $\hat{C}_1 \approx 81°0'47''$, $c_1 \approx 9{,}22$ cm; $\hat{B}_2 \approx 121°0'47''$, $\hat{C}_2 \approx 18°59'13''$, $c_2 \approx 3{,}04$ cm.

65. $\hat{A} \approx 53°7'48''$, $\hat{B} \approx 59°29'23''$, $\hat{C} \approx 67°22'48''$. $p = 21$, $S = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} =$ 84 cm^2

66. Han recorrido 27 y 36 km: $d = \sqrt{27^2 + 36^2 - 2 \cdot 27 \cdot 36\cos 70°} \approx$ 36,88 km

67. $h = x\operatorname{tg} 35° = (x + 20)\operatorname{tg} 22° \Rightarrow$ anchura $x \approx$ 27,28 m, altura $h \approx$ 19,1 m

68. Cada triángulo central tiene un ángulo de 72°: lado $= 2 \cdot 10\operatorname{sen} 36° \approx$ 11,76 cm; área $= 5 \cdot \frac{1}{2} \cdot 10^2 \operatorname{sen} 72° \approx$ 237,76 cm^2

69. $\operatorname{sen}\alpha = -\frac{2\sqrt{5}}{5}$, $\cos\alpha = -\frac{\sqrt{5}}{5}$, $\operatorname{sen}\beta = \frac{4}{5}$, $\operatorname{tg}\beta = -\frac{4}{3}$. $\operatorname{sen}(\alpha + \beta) = \frac{2\sqrt{5}}{25}$, $\cos(\alpha - \beta) = -\frac{\sqrt{5}}{5}$, $\operatorname{tg}(\alpha + \beta) = \frac{2}{11}$, $\operatorname{sen} 2\alpha = \frac{4}{5}$

70. $(\cos^2\alpha + \operatorname{sen}^2\alpha)(\cos^2\alpha - \operatorname{sen}^2\alpha) = 1 \cdot \cos 2\alpha$; $\frac{2\operatorname{sen}^2\alpha}{2\operatorname{sen}\alpha\cos\alpha} = \operatorname{tg}\alpha$

71. $2\cos^2 x - 3\cos x + 1 = 0 \Rightarrow \cos x = 1$ o $\frac{1}{2}$: $0°, 60°, 300°$

72. $\cos x(2\operatorname{sen} x - 1) = 0$: $30°, 90°, 150°, 270°$

73. Amplitud 2, período π, recorrido $[-1,3]$, desplazada $\frac{\pi}{4}$ hacia la derecha. Máximo ($y = 3$) cuando $2x - \frac{\pi}{2} = \frac{\pi}{2}$: $x = \frac{\pi}{2}$

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